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RESEARCH ARTICLE

A study of Network Attacks and Features of Secure Protocols

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ABSTRACT:

Remote protocols are essential part of the computer network systems. Since data or information is sent from one system to another, there arise many issues regarding security and the validation of this information. This paper aims to explain the basic outlook of the security measures. Every protocol which are implemented commonly are vulnerable and there are various threats that act against privacy, data integrity, availability of service, etc. It is important to have knowledge and security experiences about secure protocols. This Study contains features of the designed secure protocols. Every new protocol should meet or at least should address described issues like secure protocol does. This can be used to safeguard the existing computer networks.

KEYWORDS: Network Attacks, Secure Protocols, Security measures, Cryptography, Denial-of-Service.

INTRODUCTION:

Creating secure communication is need of the hour. In earlier days, there was difficulty in creating a network connection to utilize it in the computer program. Due to this, the programming languages developed were used in standalone or sequence machines and this was reflected in the programs of many of the compilers. The computer programming has evolved into a series of techniques. A standard computer program has functions, objects and components in which functions are the first and foremost idea. The modus operandi of object opened up the use of classes, objects and methods. In an object-oriented programming language, the three main features are the encapsulation, inheritance and polymorphism. The next stage is the use of component based and service oriented programming.

The network technologies were involved only in the component and service oriented programming. The initial networking software used network sockets and network datagrams which were prominently used in client server architecture like web, ftp or email servers. Though the use of socket programming was wide spread, it was complicated and required complex structure of codes in various products. Also in the case of a collapse of the connection or damaged packets, secure software was needed and the programmer has to remedy these circumstances with a solution code like shutting off the connection, resolving the issues and connecting again. These recovery process included personnel for writing and conversion of the binary representation by the programmer, the implantation of network protocol on application layer by software engineers. This is evidently time consuming for workforce in the network department.

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The alternate is to have a program being divided into independent parts called functions, methods, packages or components which run in independent hosts in a routine series. Few of the technologies used are RMI, RPC or CORBA. The methodology in this is that the software engineer has to create the program same as that of the standalone computers except that only certain functions of the host are connected to the network. A lower level of module is responsible for creating, closing, managing failure conditions and provides communications for it. The programmer has to work out new configuration modules for this purpose and can work in the similar manner as that of standalone computer with the help of remote resources. This method is considered to be not only economical but also the responsibility does not fall solely on the software engineer but is also shared to the network library provider.

These technologies like RPC, RMI and CORBA are binary protocols using appropriate libraries whereas SOAP or REST is text based (or XML-based) remote technologies committed on remote objects and resources. Both CORBA/RMI and REST use the references to remote objects but the former enables the programmer to revoke remote method and the latter does not require special libraries enabling the engineers to convey messages encoded in URLs and XMLs. These allow the programmer to run his program smoothly in numerous parts on the different hosts of the network simultaneously.

Though these network communication technologies are easier to work with, they relegate in the security aspects of the networking. There is a chance of every remote protocol being hacked leading to dangerous situations. Other facet of threat is that the software libraries designed without following secured set of mandates may lead to worst case scenarios like in-built security bugs like buffer overflows, indigent authentication or authorization, invalid approvals, etc. The cryptographic algorithms and the existing security systems like SSL/TLS, Kerberos or SSH are insufficient to fix these issues. Hence it is indispensable to work through and create secure remedies with configurable security techniques at par with the current security issues. This paper fundamentally emphasizes on the popular network attacks and features of secure protocol for safe communication.

Elementary weaknesses and network attacks

This part deals with the attacks that occur against the network security. The significance of the attack is difficult to comprehend as it depends on various aspects. In the case of Denial-of-service attack on a small company which does not use its web page for the main information transfer, the attack will not have much effect

whereas it is not the same in the case for an Internet company. Such attacks may demolish its reputation and image in the public. There exist few common and basic security attacks against the features of the network protocols or against its unfortified areas. Not all but few of the attacks are described here. The vectors of the upcoming attacks are evolved and improved. The basic notion of this chapter is not to describe all kinds of attacks but to classify them into categories and to identify the most threatening attacks in every category.

Physical security

The most significant feature of the security is the physical security [1] yet this is often ignored and neglected by many companies. The scholar is aware of many such physical attacks on networks where even the administrator is ignorant of it. Many physical attacks are ignored as the consequences of these are small like disruption of the network or the services offered for a short period. But devices can be stolen, data can be transferred to external devices and the point is these are often less complex than the typical network attacks. Most of the people are not aware of the threats that can happen when they use and share disks in public and in wireless network granting the access openly to their information to everybody. If the information transferred in such circumstances is confidential ones, then there is greater threat to the company. People do not worry about the computers being hacked while network surfing in the public hotspots, wireless signal being used to attack their systems and the classified information in their stolen mobiles, palmtops or laptops. The physical attacks on the computer network are not in the scope of this study and hence are not explained elaborately. However, wants to emphasize the significance of the physical threat and the necessity of the administrator to be aware of it.

Sniffing and tricking

Sniffing is one of the most basic and common methods of attacks. The concept of Ethernet is to send network frames to the entire connected device and hence the name "ether". The destination device should look for the presence of its address and if so accepts the frame transferred or ignores otherwise. The sniffing does not happen in this case as all network frames are sent to every device.

But this is not the case in some of the current networks as they are switched. Switches keep track of the network addresses and its corresponding ports. Every switch and bridge checks the source addresses in the network frames and binds them with its physical interface. Then the switch identifies the destination of the network frame and it is sent to that particular device rather than sending to all the devices. Switches divide the networks into sectors called collision domains [2]. This segregation

protects against any clashes and simple sniffing attacks and the intruder cannot easily get through this and access the data. However, since there is no authentication in the 2nd network layer in OSI/ISO model the following scenario might occur. Every device can send a frame for an instance say “my address is X” where ‘X’ is a fake physical address, the switch uses this frame binding the fake address with its physical port. Moreover, switches have very limited amount of memory capacity and too many packets with false information might cause memory overflow and the switch may start behaving as a typical hub. However, the scholar has seen switches aborting in such situations rather than working as a hub. Sniffing may also be used to diagnose the problem in many situations creating basic conflict. We want both, to have a simple diagnostic tool in network problems and to provide the network security.

Spoofing is another simple method in which since there is no authentication in the 2nd layer, making the exchange of source addresses with every device i.e. it is possible to send “ARP response frame” with wrong MAC address [3]. This is a common issue and is used ARP poisoning techniques. Since there are no states in ARP protocol, it becomes difficult to pair specific ARP request to the specific ARP response. If the host is unaware of the MAC address of the destination device, ARP requests to identify such address to every other host in the local network. The response should be given by only one of the hosts. For a clear view of the performance, the ARP responses are cached for a specific time say five minutes. If any source host has a record in the ARP cache, then another request for identification is not necessary since the destination is recognized. This paves way for the intruder to send ARP replies to the victim who usually accepts the “ARP requests”. This causes changes in the ARP table of the victim’s host. Hence the ARP cache is corrupted and is not accurate any longer. This leads to the victim sending all packets to the intruder instead of sending them to the correct destination device.

Spoofing is one of the many types of attack which may be used against Dynamic Host Configuration Protocol (DHCP). The typical context is that a new host in the network requests DHCP server for an IP addresses to use. The DHCP server allocates one of the free addresses for the requested host for a specific period and then replies to the specific MAC address. As mentioned earlier, MAC addresses can be changed and the intruder as many DHCP requests as possible one by one with varied MAC addresses. This is an easy method for Denial of DHCP Service because of no free IP addresses in the DHCP server. This leads to a situation where no one else can request and reserve a new address and connect to the network.

FEATURES OF SECURE PROTOCOL:

Code Quality

In order to make a product secure, the so-called “secure programming” principles are significant. The rules in [4], [5], [6] and [7] were specially marked by the author. The present challenge to secure protocol implementation is the lack of pen-testers which may implement such test scenarios.

All the security related issues can be fixed by checking the source code. The Find Bugs software can be used to find any vulnerability [8],[12]. The Find Bugs software does not work in the presence of “white character” in path to executable file and this was a minor issue faced during code review.

Security execution guidelines for remote protocols

The weakest part of the system is always more secured. Many systems have own communication protocols to exchange messages between hosts. We can see that it is not only significant to make the protocol secure but also to safeguard it from other dangers with the cryptography. The following secure patterns shall be implemented by the designer for any protocol.

Cryptographical Challenges

Cryptography is essential to make the Clear-text protocols safe, as anybody can listen to transmitted packets, exchange them or create new, fake packets in order to trigger some events in the system. Confidentiality has the first priority in the cryptographic algorithms as classified and sensitive messages are transmitted over the network and encryption algorithms protect such personal data as a law requirement in many countries. Consistent data has to be transmitted over the network. Distributed systems like shopping, financial operation or even in SCADA systems (Supervisory Control and Data

Acquisition) use remote protocols. The SCADA security is described in [9]. Sensitive data cannot be transacted in clear text. The protection against replay attacks shall be given by cryptographic mechanisms, which should be in-built in protocols.

By sending additional numbers in packages which are checked by the destination hosts, such mechanisms can be implemented. Random numbers as random as possible should be used for securing protocols. Weak random number generators used in applications are more vulnerable to attack. Secure software with best random generators is listed in [10], [14].

Authentication, Authorization and Accounting

Authentication process is nothing but the verification of the identity of a user or device wishing to access the application. A trust relationship is established between the user or other service and the application by this process. Authentications are of different types. Server authentication is required if the client wants to trust the server. Users who want to perform some operations on the other site are authenticated by almost every server. Secure protocol provides mutual authentication which is the best way of authentication, where both sides have to authenticate each other. There are few types of authentication mechanisms. "Something you know" example, passwords, PIN codes etc. are the first type which is familiar among us.

Passwords are like users' identities with this approach and it should not be shared with anybody. Second type is "something you have" where it represents things like magnetic cards, smart cards, software and hardware tokens and many others. The third group is confirmed by "something you look like", which is a person's identity, for example fingerprint or iris in the eye. Fourth group of checking identity is "something you are" which represents things like handwritten signature or tone of the voice though these methods is same as that in third group as both includes biometric methods. The last group is behavior-related and the third group contains static properties is the difference between the two. At least two groups of authentications should be used as mentioned by many of the guidelines. If two or more types of authentication are used, then it would be much more difficult for the intruder, to give the correct credentials. Authentication process alone is not sufficient [11], [13]. This where authorization comes in, where We have to check user's right before we perform requested operation as every legitimate user have their own right in the system. Guest users have rigid rights than trusted users and devices who may have permissive rights. If a secure system is needed, appropriate configuration and user management have to be followed. Excessive privileges in the system lead to perform unnecessary actions by too many users. Access control is of few types. First type is Mandatory Access Control (MAC) which is based on the sensitivity of the resource. Discretionary Access Control (DAC) works on the idea that the owner of the resource has the ability to change the permissions for rest of users/devices which is widely used control. When predefined rules decide whether the access is granted or denied, it is called Rule-Based Access Control (RBAC) which is yet another method of authorization. Last method is Role-Based Access Control (RBAC). Although the acronym is the same, the concept is quite different where Role-Based Access Control is related to user's roles in the organization. The access is granted depending upon the role which allows for such access to

specified resource as every user/device has at least one role in the organization. Detailed information about described access control methods is in [12]. Based on the organization's requirements, the software designer may use default authorization mechanism or may take full control and authorize other parts of the system in secure protocol proposition.

Logging the performed operations is related to accounting. Any operation may be performed only if the user is authenticated and authorization allows for performing specified operation and they should be logged in case of any security issues. In the instance of any investigation and legalities, the logs from potential security incident are helpful. In remote protocols Authentication, Authorization and Accounting are necessary. The designer shall integrate these mechanism and additional implementation may be not enough because of initial security issues in original library with protocol.

Denial-of-service

The Denial-of-Service (DoS) attacks are very difficult to prevent. When huge numbers of similar requests appear, then localized DoS attacks may be discovered and the source of the attack may be blocked and the server may be freed from unnecessary operations. However these types of attacks are simple and does not happen often as every IDS (Intrusion Detection System) may discover and protect against such attacks. But the complicated one is the Distributed DoS as the source of attack is varied and distributed. Only with the help of very good IDS and additional network connection such attacks can be prevented and the system can be protected.

The administrator should manage the limits for specified operations in a secure protocol. It is very hard to implement operations like blocking the source of attack and some part of the risk may be lessened with the aid of appropriate firewall and IDS. The current secure protocol implementation has yet to solve this kind of attack.

CONCLUSION:

Protecting the computer networks is consequential. There is no single technique to protect data in networks as the issues or threats to the data in networking are not only concerned about data integrity and confidentiality but is varied, vast and much more complex. The science of Cryptography is essential and a must, but is not limited to it, as the troubles are not only about cryptanalysis but about threats related to social engineering, denial of service or software vulnerabilities like buffer overflows, SQL injections, cross site scripting, etc those of which will be seen in the next chapter. The solutions for these kinds of threats are provided by hundreds of clear-text

protocols and the cryptographic algorithms are discussed in chapter three.

The range of probable targets of networks is incredible. Any and every device, data and user can become a target. It does not matter whether the device or data is of any significance. Sometimes unused and bygone devices become a target can be used in illegal or criminal activities in local network or can be used as a component of the system. An added advantage is that these devices might not be patched and the hacker could have easy access to these or their downloaded exploits. The intruders can use data that are commonly left as insignificant to their advantage to gain the users' confidence and acquire more relevant and personal information about the user.

Many of the users of network protocols depend upon other protocols which includes the intruders also. There is no foolproof universal security model of computer networks. Those of the existing protocols also have to use other protocols as they are designed for specific requirements only.

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